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PAPER

Imaging of vaporised sub-micron phase change contrast agents with high frame rate ultrasound and optics

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9 March 2018**Abstract**

Phase-change ultrasound contrast agent (PCCA), or nanodroplet, shows promise as an alternative to the conventional microbubble agent over a wide range of diagnostic applications. Meanwhile, high-frame-rate (HFR) ultrasound imaging with microbubbles enables unprecedented temporal resolution compared to traditional contrast-enhanced ultrasound imaging. The combination of HFR ultrasound imaging and PCCAs can offer the opportunity to observe and better understand PCCA behaviour after vaporisation captures the fast phenomenon at a high temporal resolution. In this study, we utilised HFR ultrasound at frame rates in the kilohertz range (5–20 kHz) to image native and size-selected PCCA populations immediately after vaporisation *in vitro* within clinical acoustic parameters. The size-selected PCCAs through filtration are shown to preserve a sub-micron-sized (mean diameter < 200 nm) population without micron-sized outliers (>1 μm) that originate from native PCCA emulsion. The results demonstrate imaging signals with different amplitudes and temporal features compared to that of microbubbles. Compared with the microbubbles, both the B-mode and pulse-inversion (PI) signals from the vaporised PCCA populations were reduced significantly in the first tens of milliseconds, while only the B-mode signals from the PCCAs were recovered during the next 400 ms, suggesting significant changes to the size distribution of the PCCAs after vaporisation. It is also shown that such recovery in signal over time is not evident when using size-selective PCCAs. Furthermore, it was found that signals from the vaporised PCCA populations are affected by the amplitude and frame rate of the HFR ultrasound imaging. Using high-speed optical camera observation (30 kHz), we observed a change in particle size in the vaporised PCCA populations exposed to the HFR ultrasound imaging pulses. These findings can further the understanding of PCCA behaviour under HFR ultrasound imaging.

1. Introduction

High-frame-rate (also known as ‘ultrafast’) contrast-enhanced ultrasound imaging (HFR-CEUS), by combining microbubble (MB) contrast agents with significantly improved temporal resolution, offers new opportunities for improved imaging and quantification of blood flow and perfusion (Tanter and Fink 2014, Tremblay-Darveau *et al* 2014, Viti *et al* 2016). Couture *et al* (2012) demonstrated the benefits of using plane-wave imaging for significantly increasing the contrast without MB disruption compared with conventional contrast imaging. For example, HFR ultrasound imaging with MBs is capable of tracking and quantifying fast and dynamic flows providing diagnostic information for vascular diseases such as atherosclerosis (Leow *et al* 2015). Compared to conventional Doppler or imaging at normal clinical frame rates, HFR with MBs excels at temporal resolution over a full field of view in case of fast changing/moving targets (e.g. heart or arterial flow), or alternatively can provide better signal-to-noise ratio by averaging the extra temporal samples acquired using HFR. Acoustic

super-resolution and blood velocity mapping can be also realised, breaking the fundamental diffraction limit of ultrasound waves (Errico *et al* 2015, Christensen-Jeffries *et al* 2017). Image contrast in micro-vascular imaging can also be significantly improved through signal processing of HFR-CEUS data (Stanziola and Tang 2017).

As one of the most actively researched alternative ultrasound contrast agent, the sub-micron phase-change contrast agent (PCCA), or nanodroplet, has shown attractive advantages and new perspectives in ultrasound imaging and therapy (Kripfgans *et al* 2000, Kawabata *et al* 2005, Rapoport 2012, Reznik *et al* 2012b, Sheeran and Dayton 2012). The PCCAs, with the sub-micron initial size, can potentially migrate and accumulate into extravascular space within a tumour, where they can be acoustically activated/vaporised into echogenic MBs via phase transition offering ultrasound contrast enhancement *in situ*. This normally requires a specific imaging pulse sequence since the liquid-cored PCCAs cannot provide a contrast until turning into MBs via vaporisation. For example, the ‘imaging-activation-imaging’ pulse sequence allows both acoustic activation and imaging of the vaporised PCCAs, typically implemented on a programmable ultrasound research platform (Puett *et al* 2014, Sheeran *et al* 2015, Lin *et al* 2016). It has also been demonstrated on clinical and pre-clinical commercial ultrasound scanners by adjusting the acoustic energy output (Choudhury *et al* 2017, Sheeran *et al* 2017a).

The existing literature has yet to evaluate HFR ultrasound imaging with vaporised sub-micron PCCAs, and the behaviour of the vaporised PCCAs exposed to HFR ultrasound imaging pulses. There are a few studies that have reported the acoustic and optical measurements of vaporised sub-micron PCCAs, for instance, Reznik *et al* (2011) characterised fluorosurfactant encapsulated PCCAs using a single-element transducer transmitting ten interrogation pulses over 1 s post-activation, while Ishijima *et al* (2016) optically investigated the lifetime of PCCAs. There are also other studies on microdroplets. For example, Couture *et al* (2011) applied 3200 Hz ultrafast ultrasound for tracking the release of microdroplets for an ultrasound internal tattooing purpose. Kang *et al* (2014) investigated bubble generation from phospholipid-encapsulated microdroplets using an acoustic activation pressure beyond the FDA-stipulated limit for diagnostic ultrasound. Therefore, how the vaporised sub-micron PCCAs respond to HFR-CEUS plane-wave imaging pulses has not been studied.

Herein, we present, to the best of our knowledge, the first investigation into HFR-CEUS with clinically relevant parameters to image vaporised lipid-shelled, perfluorocarbon-cored PCCA populations in a 200 μm -cellulose tube phantom with simultaneous high-speed (30 kHz) microscopic observation in the course of 500 ms after acoustic activation pulse. This work is a continuation of a previous study (Lin *et al* 2017a) presented at the 2017 IEEE International Ultrasonics Symposium. This will provide an improved understanding of the imaging contrast enhancement from the vaporised PCCAs with HFR-CEUS, which has potential importance for developing PCCA-specific imaging strategies.

2. Methods

2.1. Precursor MB and PCCA preparation

The sub-micron lipid-shelled perfluorocarbon PCCA emulsion was made in-house by condensing the precursor MBs using the protocol described in Sheeran *et al* (2011, 2016b). Briefly, the lipid solution was firstly prepared by dissolving 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC), 1,2-distearoyl-sn-glycero-3-phosphoethanolamine-N-[methoxy(polyethylene glycol)-2000] (18:0 PEG2000 PE) 9:1, m:m (total lipid concentration of 0.81 mg ml^{-1}) in propylene glycol, glycerol, and phosphate-buffered saline (15/5/8, v/v/v). The lipids were purchased from Avanti Polar Lipids, Inc., USA. One ml of this excipient lipid solution was sealed in a 2 ml glass vial with a headspace of decafluorobutane vapor (Fluoromed, USA). Next, the precursor MB was produced by standard mechanical agitation (Vialmix, USA). Finally, the PCCA emulsion was generated via condensing precursor MBs in a $-10\text{ }^{\circ}\text{C}$ bath environment. In the experiment, the acoustic response of the precursor MBs were used as a control to highlight the different behaviours between the vaporised PCCAs under HFR ultrasound imaging.

It is crucial to control the size of the vaporised PCCA population for different diagnostic and therapeutic applications. Previous studies have identified and quantified the precursor MB outliers (Sheeran *et al* 2016a) before condensation using a similar formulation as that used in this study. In order to remove the size outliers (diameter $>1\text{ }\mu\text{m}$) in the stock native PCCA emulsion, the size-selected PCCAs were produced by mechanical filtration, where the native PCCA emulsion was gently mechanically filtered by Acrodisc syringe filters with a 1 μm PTFE membrane (Pall Corp, USA) at an approximate rate of 2 ml min^{-1} following the protocols according to Cheung *et al* (2008). To minimise clogging of the filter pores, the filter with the largest filtration area (3.9 cm^2 , 25 mm in diameter) commercially available was utilised in this study, and the filter was discarded after filtering 1 ml of native PCCA emulsion. The native, size-selected PCCAs, and precursor MB solution was diluted to 0.05% relative to their original concentrations for the main ultrasound and optical experiment except stated otherwise. Although there is some guidance on the dosage for *in vivo* studies specially for this type of sub-micron PCCA (Sheeran *et al* 2015), the PCCA concentration used in this study was at the same level ($\sim 10^7$ particles ml^{-1}) as that used in existing *in vitro* experimental research using a similar type of PCCA (Sheeran *et al* 2016a, 2017a), which

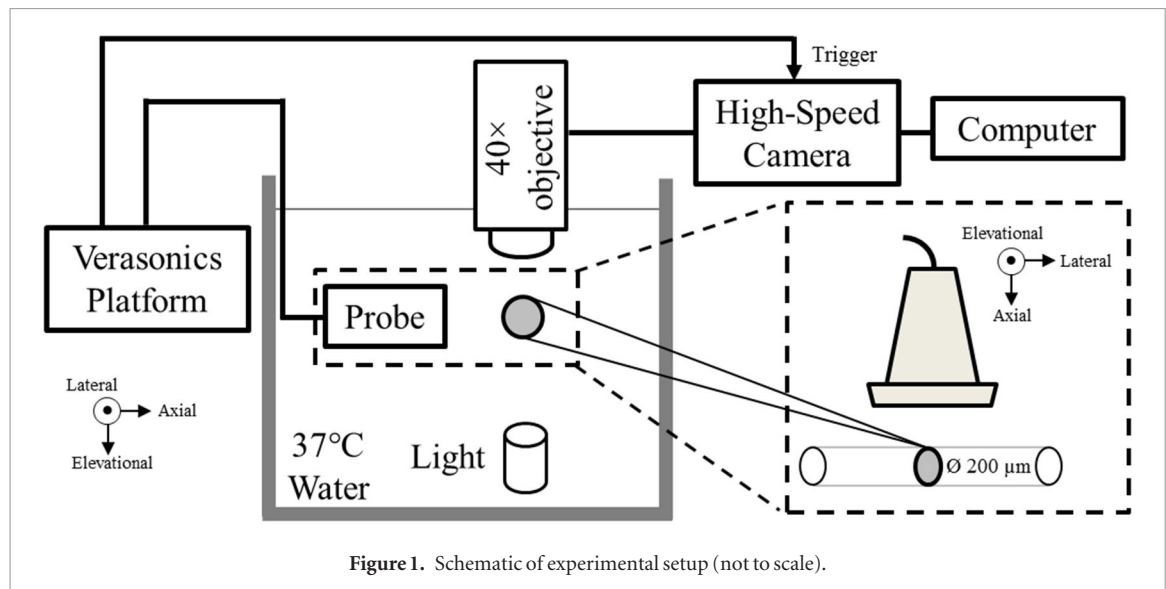


Figure 1. Schematic of experimental setup (not to scale).

also falls in the ‘linear range’ (Li *et al* 2015) such that the acoustic and optical quantification behave linearly with the PCCA concentration. Further discussion on the concentration is provided in section 4.

2.2. Characterisation of MB and PCCAs

The size and concentration of native/size-selected PCCAs were measured by a NanoSight NS 300 instrument (Malvern Instruments, UK) following the guidelines (Hackley and Clogston 2011), in which the particle size measurement range was 10–1000 nm. The native/size-selected PCCA samples were diluted 1/1000 in purified and deionised water (milliQ, Canada) before which the stock emulsion was well mixed to avoid ‘large-droplet’ sedimentation (Sheeran *et al* 2016a). All the measurements were conducted applying identical settings (i.e. Camera Level 15, Detect Threshold 5). Each sizing and counting result was produced by an average of three repeated measurements (1498 frames acquired for each measurement) per sample over four samples (12 measurements in total). The precursor MBs were sized and counted using the approach detailed in (Sennoga *et al* 2010).

2.3. Experimental setup

The contrast agents were imaged in a microvessel phantom, which was an optically and acoustically transparent 200 μm (inner diameter) cellulose tube (Cuprophane, RC55 8/200 Membrana GmbH, Germany) submerged in 37 °C water that reached gas equilibrium (Mulvana *et al* 2012) before experiment (figure 1). The programmable ultrasound research platform (Vantage 128, Verasonics, USA) equipped with a clinical L11-4 linear array (Verasonics, USA) was employed to deliver the ‘image-activation-image’ pulse sequence for activating and imaging the PCCAs. The probe was submerged in the water tank, and positioned axially 15 mm away from the tube.

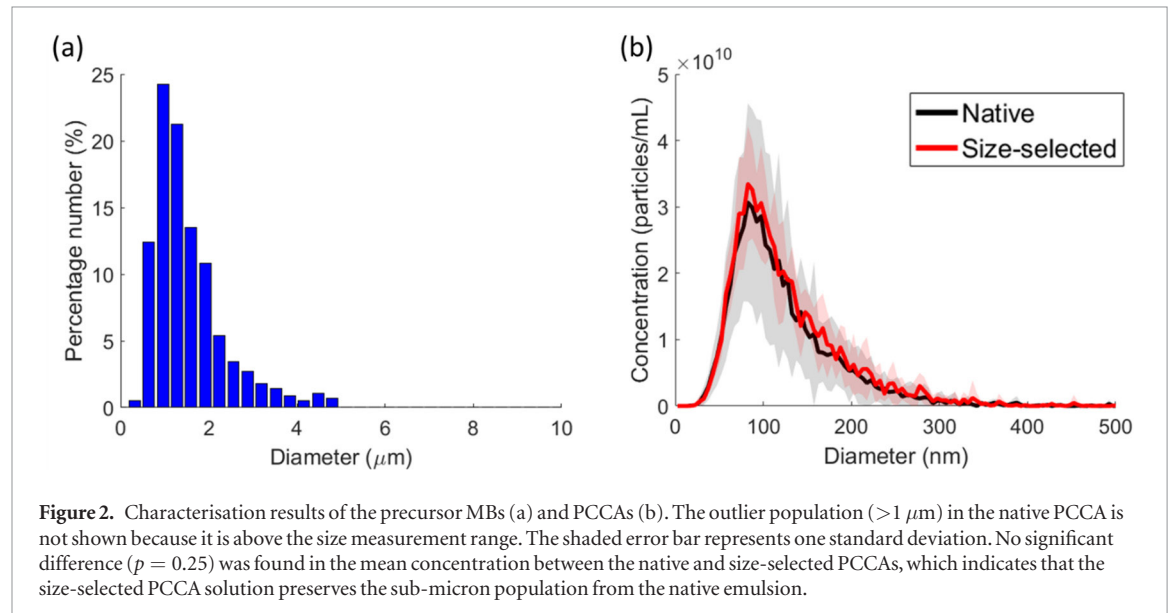
The optical observation was achieved by a water immersion objective lens (Olympus, LUMPlanFL, 40×, NA = 0.8) connected to a high-speed camera (FASTCAM SA5, Photron, USA), which was triggered by a Verasonics system at the time the second imaging pulse sequence initiated. The optical imaging was used to observe the behaviour of vaporised PCCAs as a result of each HFR ultrasound imaging pulses. The optical system was calibrated using a reticle. The 40× objective lens resulted in a nominal resolution of 0.5 μm per imaging pixel. The camera was triggered to operate at 30 k frames s⁻¹. The optical imaging frame rate was set such that there was at least one optical image frame between any neighbouring ultrasound pulses. The optical imaging plane was focused and kept at the middle-plane of the tube throughout the experiment. During experiment, the PCCAs were slowly administered into the tube, followed by clamping two ends of the tube to stop the flow inside, and waiting for 15 s prior to the onset of the whole pulse sequence. This process was repeated for each experiment’s data acquisition with replenishment of the fresh contrast agents into the tube. The acoustic absorbers were used to line the water tank to reduce ultrasound reflections.

2.4. Ultrasound parameters

The applied acoustical parameters are detailed in table 1. The ultrasound settings were based on a previous *in vitro* experiment (Lin *et al* 2017b) of imaging with the same type of PCCA. In the imaging state, for both ‘before’ and ‘after’ activation, pulse-inversion (PI) contrast-specific imaging was achieved by transmitting a pair of phase-inversed 4 MHz single-cycle single-angle (0°) plane-waves with a range of low pressures, i.e. mechanical index (MI) 0.05, 0.07, and 0.1, at various pulse-repetition-frequency (PRF) HFR: 5, 10, and 20 kHz and LFR:

Table 1. ‘Imaging-activation-imaging’ sequence and ultrasound parameters for PCCA activation and imaging.

State	Pulse type	Transmit frequency (MHz)	Pulse length (cycle)	MI	PRF (Hz)
Imaging (before)	Plane-wave	4	1	0.05/0.07/0.1	5 k/10 k/20 k/50
Activation	Focus	8	5	1.70	13.7 k
Imaging (after)	Plane-wave	4	1	0.05/0.07/0.1	5 k/10 k/20 k/50



50 Hz. Single-angled plane-wave imaging was applied in this study to avoid the ‘averaging effect’ from multi-angled images such that the acoustic response to every imaging pulse could be captured. In the ‘activation’ state, the PCCAs in the tube were activated by electronically steering focused waves (8 MHz, 5-cycle, $\text{MI} = 1.70$, $f\text{-number} = 0.4$, $\text{PRF} = 13.7\text{kHz}$) at 15 mm away from the probe surface. The activation of the frequency was selected to be 8 MHz to improve the ‘superharmonic focusing’ effect (Shpak *et al* 2014) maximising the vaporisation probability. For the control experiment, the precursor MBs were imaged with the identical parameters as those used for imaging the vaporised PCCAs without the activation pulse. The pressure field of both the plane-wave and focus-wave were calibrated using a 0.2 mm needle hydrophone (Precision Acoustics, UK).

2.5. Ultrasound image and data analysis

Both ‘positive’ and ‘negative’ plane-wave data corresponding to a pair of PI transmission were collected, based on which both the PI and B-mode images were formed for off-line image analysis. A PI image was formed by summing a pair of ‘positive’ and ‘negative’ frames, while a B-mode image was just a ‘positive’ frame (equivalent to a ‘negative’ frame). Differential images were generated using pixel-based subtraction of the baseline image (i.e. ‘before’ image) from the images after activation (i.e. ‘after’ image). The image pixel amplitude of these differential images was quantified in the selected region-of-interest (ROI) at the tube region, and was then normalised to the maximum value of all the datasets presented in the same figure and plotted as a function of time. Each result was presented by the mean, and the shaded error bar showing plus and minus one standard deviation over six independently repeated experiments. The Student’s t -test was used to compare the statistical difference between groups with $p < 0.05$ considered to be significantly different.

3. Results

3.1. Characterisation of the contrast agents

The number percentage distribution of the precursor MBs (figure 2(a)) shows a mean diameter of $1.22 \pm 0.43\ \mu\text{m}$, a median of $1.16\ \mu\text{m}$, and a mode of $0.96\ \mu\text{m}$. The results of the characterisation of the PCCAs show the average diameters as follows: mean $118.7 \pm 6.7\text{ nm}$ and mode $84.4 \pm 7.7\text{ nm}$ for native PCCAs, and mean $126.4 \pm 5.7\text{ nm}$ and mode $83.7 \pm 4.4\text{ nm}$ for size-selected PCCAs. The concentration distribution of both the native and size-selected PCCAs is presented in figure 2(b), with mean concentration (in particles/ml) $3.31 \times 10^{10} \pm 0.3 \times 10^{10}$ for native PCCA and $3.45 \times 10^{10} \pm 0.13 \times 10^{10}$ for the size-selected PCCA emulsion. There is no significant difference in the mean concentration between the native and size-selected PCCAs. This

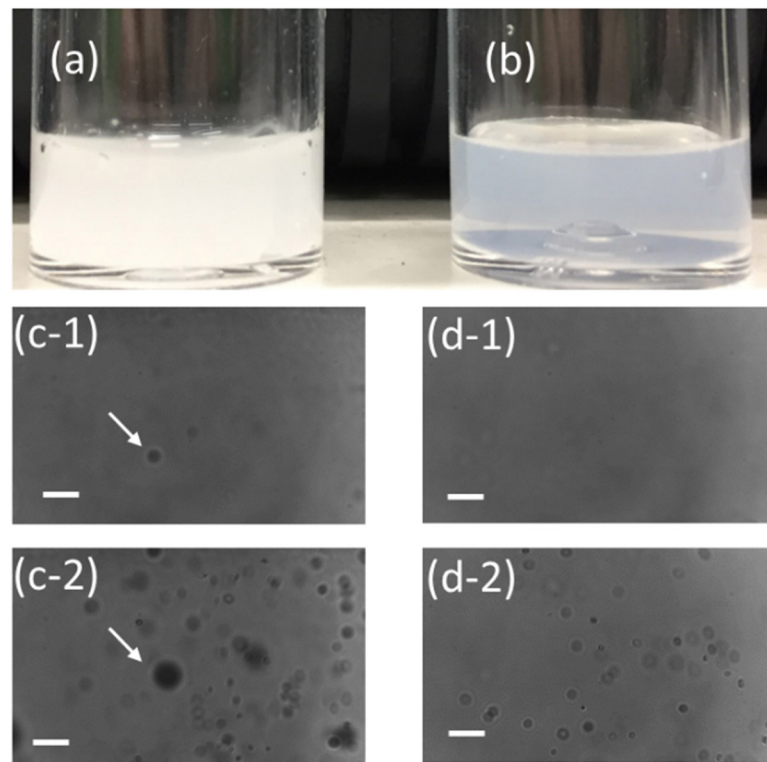


Figure 3. Appearance of the native (a) and size-selected (b) PCCAs, representative microscopic images of 3% diluted (relative to stock concentration) native PCCAs before (c-1) and after (c-2) acoustic activation, size-selected PCCAs before (d-1) and after (d-2) acoustic vaporisation. The white arrows indicate the size of the outlier in the native PCCAs before and after vaporisation. The scale bar is 15 μm . No PCCA outliers were observed from (d-1), which suggests the successful outlier-removal via the filtration process.

indicates that the removal of the outliers does not significantly influence the sub-micron population and its size distribution. Due to the measurement limit of the NanoSight instrument we used, the PCCA outliers (diameter $>1\ \mu\text{m}$) were not measured.

After the filtration of native PCCA emulsion (figure 3(a)), the appearance of the size-selected PCCA emulsion (figure 3(b)) became more translucent. Prior to vaporisation (figure 3(c)), a PCCA outlier ($>1\ \mu\text{m}$, white arrow) could be visualised on the microscopic images (c-1) of the native PCCAs, while most of the size-selected PCCAs (d-1) were beyond the resolution of the optical imaging system. After acoustic activation (figure 3(d)), the vaporised PCCAs became visible in the same field-of-view. The vaporised PCCA outliers (newly created ‘large’ MBs, $>10\ \mu\text{m}$) were generated (c-2) out of the PCCA outliers, whereas the majority of the vaporised size-selected PCCAs were under $10\ \mu\text{m}$, which were notably smaller and more uniform in size distribution.

3.2. Ultrasound imaging and quantification

Figure 4 shows representative PI (a)–(d) and B-mode (e)–(h) images for the native PCCAs in a 200 μm cellulose tube at different time points after activation pulse. Before activation, the background signal was removed by differential imaging subtraction for both the PI (figure 4(a)) and B-mode (figure 4(e)) images. In the PI images, a significant contrast enhancement was generated within the tube region immediately after the activation pulse (figure 4(b)), and then the signal amplitude decreased afterwards (figures 4(c) and (d)). The B-mode also exhibited an evident increase in contrast (figure 4(f)) immediately after the activation, and then a decreasing trend (figure 4(g)) in the first time period. However, there was a notable recover of the signal amplitude (figure 4(h)) afterwards.

Similarly, the vaporised size-selected PCCAs (figure 5) generated significant contrast enhancement (figures 5(b) and (f)) after the activation pulse. Then the signal amplitude decreased for both the PI (figures 5(c) and (d)) and B-mode (figures 5(g)–(h)) images. However, no signal was recovered for the B-mode imaging of the size-selected PCCAs (figures 5(e)–(h)).

The acoustic responses of the vaporised PCCAs and MBs with HFR ultrasound imaging as well as the vaporised PCCAs with LFR ultrasound imaging as shown in figures 6 and 7 over 500 ms after vaporisation with details (0–20 ms) in their sub-figures. The backscattered signal amplitude of precursor MBs (magenta curves) persisted over the entire imaging acquisition period of time and with the same experimental parameters, that of the vaporised PCCAs (all the blue curves) show contrast generation immediately after the PCCA activation (at 0 ms) followed by signal decay in the first tens of milliseconds, and the decreasing trend fit to linear regression

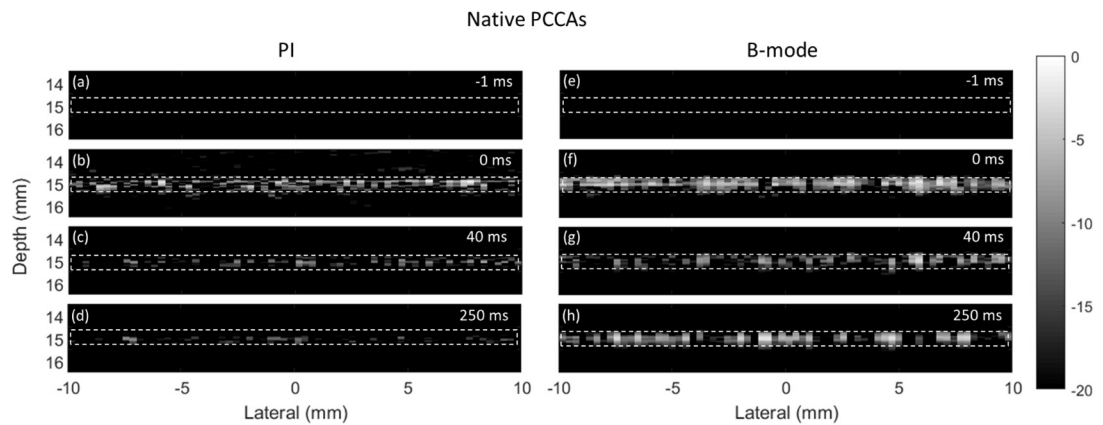


Figure 4. Representative PI (a)–(d) and B-mode (e)–(h) ultrasound images (acquired at 10 kHz, $MI = 0.07$) of the native PCCAs in a 200 μm cellulose tube. The times after activation for these selected PI images are (a) –1 ms, (b) 0 ms, (c) 40 ms, and (d) 250 ms; for the B-mode images (e) –1 ms, (f) 0 ms, (g) 40 ms, and (h) 250 ms. The dashed white boxes indicate the tube regions, and are the ROIs for quantitative analysis. All the images are shown in the 20 dB range.

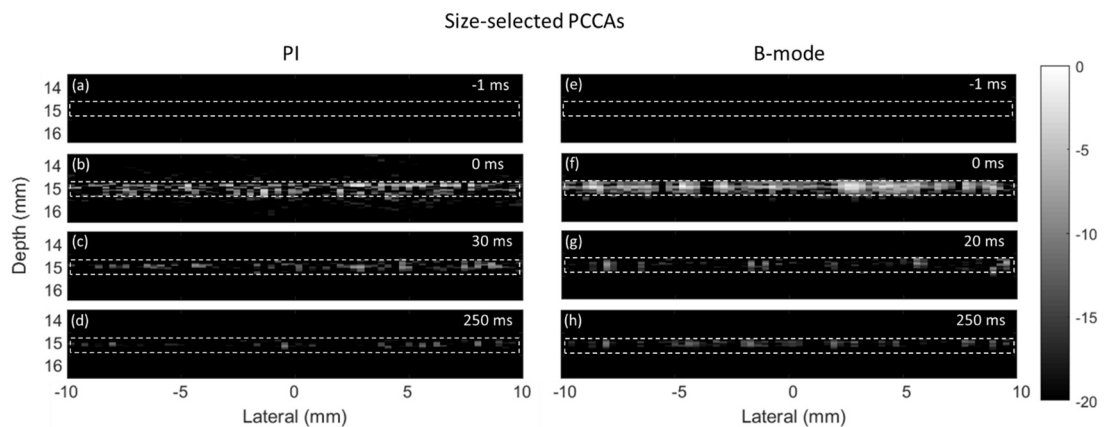


Figure 5. Representative PI (a)–(d) and B-mode (e)–(h) ultrasound images (acquired at 10 kHz, $MI = 0.07$) of the size-selected PCCAs in a 200 μm cellulose tube. The times after activation for these selected PI images are (a) –1 ms, (b) 0 ms, (c) 30 ms, and (d) 250 ms; for the B-mode images (e) –1 ms, (f) 0 ms, (g) 20 ms, and (h) 250 ms. The dashed white boxes indicate the tube regions, and are the ROIs for quantitative analysis. All the images are shown in the 20 dB range.

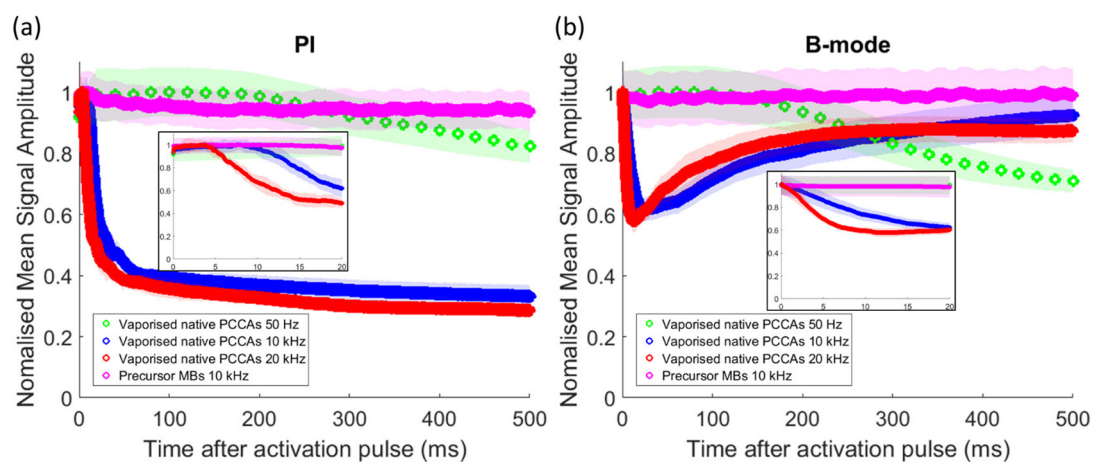
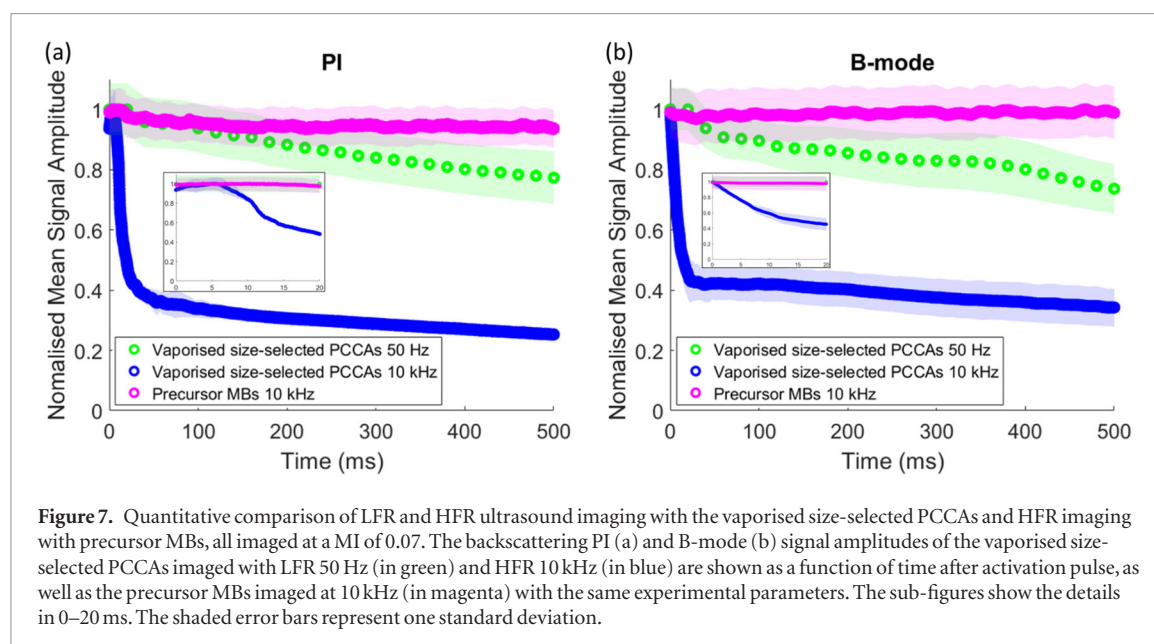


Figure 6. Quantitative comparison of LFR and HFR ultrasound imaging of the vaporised native PCCAs and HFR imaging with precursor MBs, both imaged at a MI of 0.07. The backscattering PI (a) and B-mode (b) signal amplitudes of the vaporised native PCCAs imaged with LFR 50 Hz (in green), HFR 10 kHz (in blue), and HFR 20 kHz (in red) were shown as a function of time after activation pulse, as well as the precursor MBs imaged at 10 kHz (in magenta) with the same experimental parameters. The sub-figures show the details in 0–20 ms. The shaded error bars represent one standard deviation.



(least-squares fit, coefficient of correlation $R^2 \geq 0.95$) showing gradient coefficients of -0.03 (0–20 ms, figure 6(a), red), -0.014 (0–40 ms, figure 6(a), blue), -0.027 (0–17 ms, figure 6(b), red), -0.196 (0–20 ms, figure 6(b), blue), -0.025 (0–30 ms, figure 7(a)), -0.027 (0–20 ms, figure 7(b)). The effect of the frame-rate appears in the first tens of milliseconds: the higher the frame rates, the faster the signals decay. The B-mode signal from the vaporised native PCCAs turned into a growth trend after that initial decrease, which was significantly different from that of the vaporised size-selected PCCAs. This difference was primarily due to the presence of the newly produced MB outliers generated from the PCCA outliers ($>1 \mu\text{m}$). With LFR ultrasound imaging (green curves), both the PI and B-mode signal amplitudes of the vaporised native/size-selected PCCAs illustrated a small decay, reaching approximately 0.8 at 500 ms.

3.3. Optical imaging of the vaporised PCCAs

Figures 8(a)–(f) demonstrate the optical images of the vaporised native PCCAs acquired at selected time points after acoustic vaporisation, and figures 8(i) and (ii) show the detailed phenomena in the selected ROIs in figure 8(b) (see supplementary data S1 (stacks.iop.org/PMB/63/065002/mmedia), video playbacks at 50 Hz). Prior to activation, the majority of the PCCA populations was under the resolution of the optical imaging system (figure 8(a)). After activation, the vaporised PCCAs were generated in the range of $1\text{--}10 \mu\text{m}$, e.g. shown in ROI (i) with a size outlier of approximately $12 \mu\text{m}$, e.g. presented in the ROI (ii) activated from a PCCA outlier. In the first ten milliseconds, apparent translation and coalescence phenomena were demonstrated for both the normal-sized MBs (figure 8(i)) and outlier MBs (figure 8(ii)). In figure 8(i), the first coalescence event was observed for the two vaporised PCCAs (off the optical focus plane), and lasted for approximately 0.9 ms. The second event for the two large vaporised PCCAs distinctively showed the following stages: collision, flattening of contact surfaces, liquid film drainage and rupture, and final formation in the course of approximately 6.57 ms, similar to a previous study on MB coalescence (Postema *et al* 2004). In the later phase (figures 8(d)–(f)), the translation of the vaporised PCCAs in the ultrasound propagation direction (towards the distal tube wall), was the most visible phenomenon due to the primary acoustic radiation force.

The control optical experiment was conducted with the vaporised PCCAs (figure 9, also see supplementary data S2, video playbacks at 50 Hz) by operating the same optical observation parameters after the activation pulse, without the presence of the ultrasound field. Without the acoustic interrogation, the vaporised PCCA populations remained relatively stationary throughout the entire time period, and no sign of coalescence was observed even when some mutual distances were close ($<20 \mu\text{m}$).

3.4. Effects of the MI

To investigate the effects of the imaging MI on the vaporised PCCAs, the temporal evolution of the backscattering signal for the vaporised native PCCAs was recorded with various acoustic pressures (MI = 0.1, 0.07, and 0.05) at a fixed imaging frame rate (10 kHz), as shown in figure 10. For the PI signals, as the MI decreased, the decreasing rate of the signal became lower in the first tens of milliseconds, with the gradient coefficients (in ms^{-1}) of -0.028 (0–25 ms), -0.014 (0–40 ms), and -0.0065 (0–100 ms) for MIs of 0.1, 0.07, and 0.05, respectively, reaching a plateau of relative amplitude at 0.4 afterwards. For the B-mode, the signal decayed in a similar fashion in the first tens of milliseconds for the PI mode, following a linear decreasing trend with gradient coefficients (in ms^{-1})

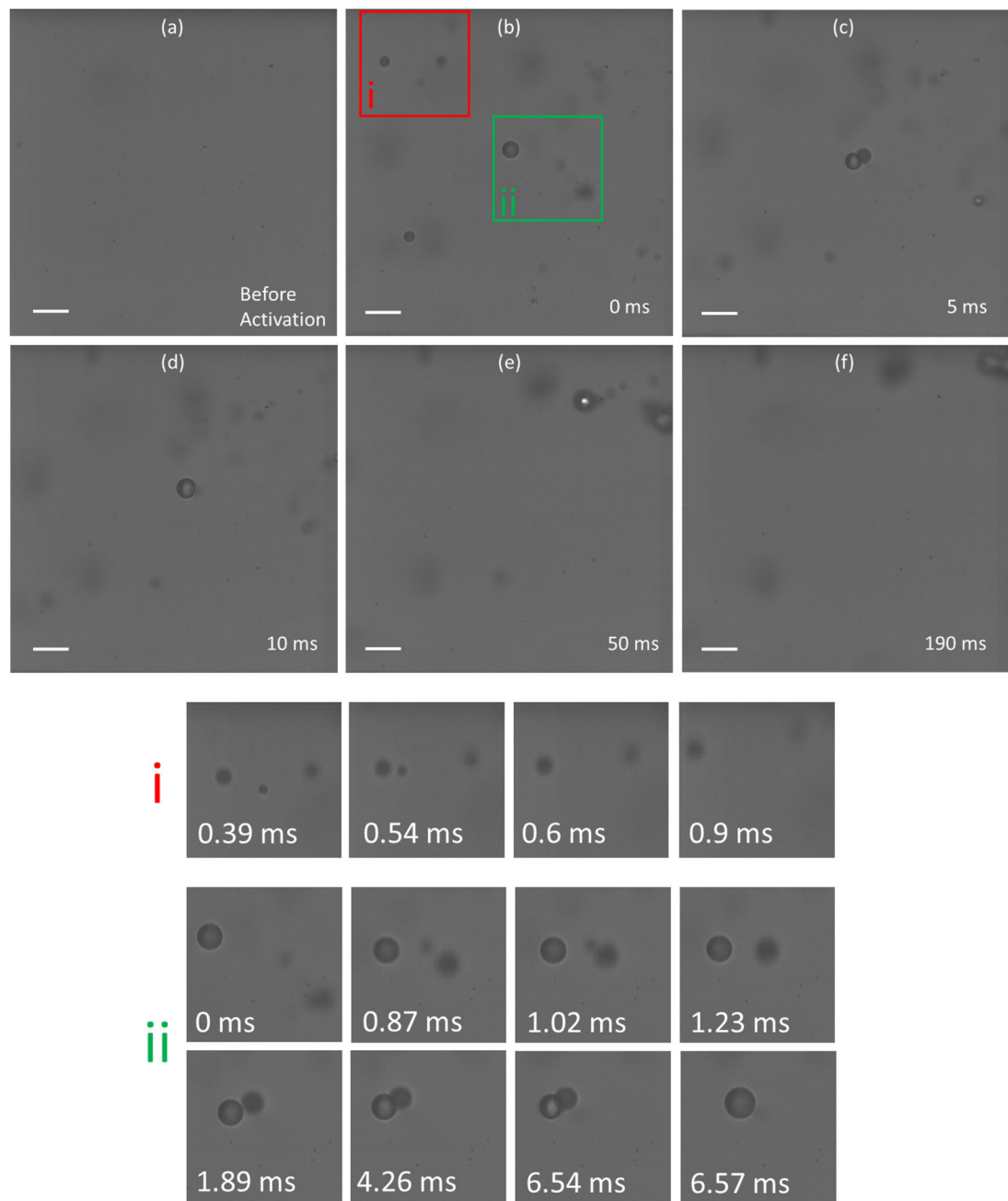


Figure 8. (a)–(f) Representative optical images of native PCCAs acquired at 30 kHz (a) before activation, (b) 0 ms, (c) 5 ms, (d) 10 ms, (e) 50 ms, and (f) 190 ms after the activation pulse. The vaporised native PCCAs were simultaneously driven by 20 kHz, 0.1 MI HFR ultrasound imaging pulses. The scale bar is 20 μm . (i) Selected ROI showing one coalescence event with normal-sized vaporised PCCAs. (ii) Selected ROI showing one coalescence with large-sized vaporised PCCAs. Each image corresponds to a $61 \times 61 \mu\text{m}^2$ area.

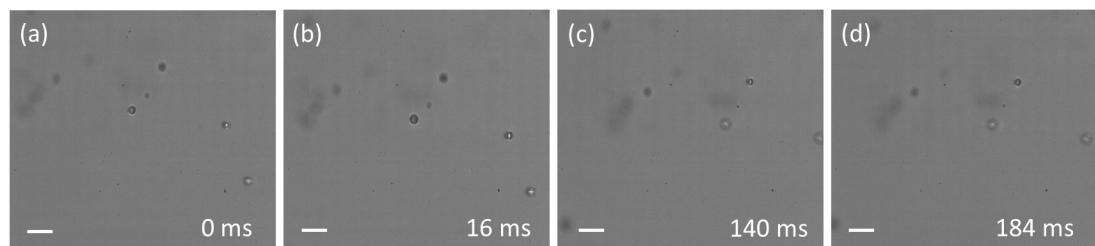
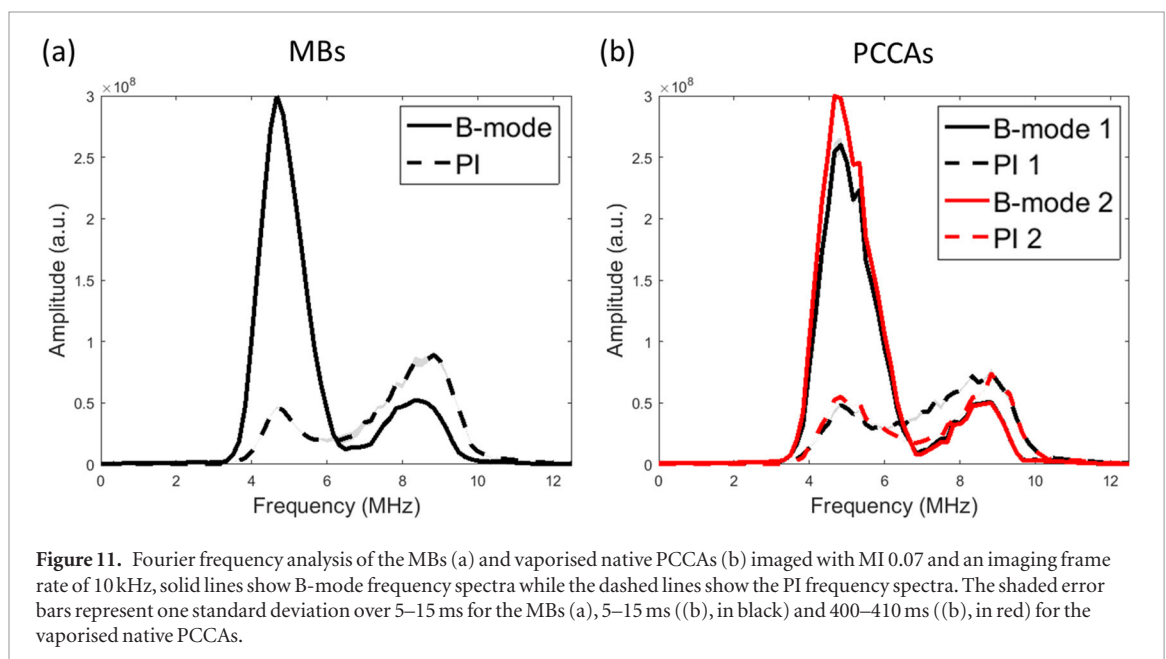
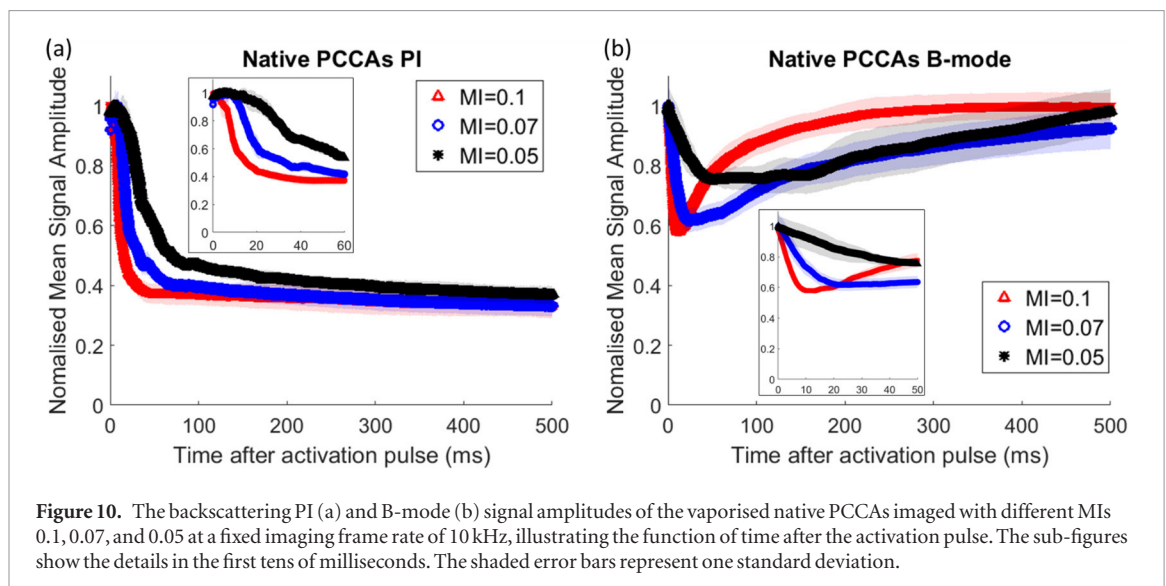


Figure 9. Representative optical images of the vaporised size-selected PCCA populations acquired at 30 kHz (a) 0 ms, (b) 16 ms, (c) 140 ms, and (d) 184 ms after the same activation pulse, in the absence of ultrasound imaging interrogation. The scale bar is 20 μm .



of -0.042 (0–10 ms), -0.0196 (0–20 ms), and -0.0049 (0–50 ms) for MIs of 0.1, 0.07, and 0.05, respectively. Afterwards, the B-mode signal amplitude returned to the original amplitude level. Therefore, it indicated the effect of acoustic pressure on the signal decay in the first tens of milliseconds: the higher the acoustic pressures, the faster the signals decreased.

3.5. Frequency spectrum analysis

A representative Fourier frequency analysis of the MBs and vaporised native PCCAs imaged at a 0.07 MI and a 10 kHz frame rate is shown in figure 11. The PI technique allows the suppression of fundamental signals while preserving the even harmonic signals with the bandwidth of the probe. Both the MB and vaporised PCCA show similarities in the spectra of the B-mode and PI. Over the entire imaging time (0–500 ms), the frequency spectrum of the MB remained the same as shown in figure 11(a). The fundamental frequency component of the vaporised native PCCAs increased, while the second harmonics decreased in the later stage (400–410 ms, figure 11(b), red) after the early stage (5–15 ms, figure 11(b), black).

4. Discussion

In this study, the acoustic responses and behaviours of vaporised native/size-selected PCCA populations were studied taking advantage of the high temporal resolution of both the HFR ultrasound and optical imaging approaches. A significant difference in the acoustic signals between the vaporised PCCAs and MBs with HFR ultrasound imaging has been shown. After acoustic vaporisation, the harmonic ultrasound signal (PI) of both

the vaporised native and size-selected PCCAs decreased in the first tens of milliseconds, and the fundamental signal (B-mode) of the native PCCAs recovered afterwards, but with no recovery of the size-selected PCCAs. For the MBs, there was no sign of signal decay under the same HFR ultrasound imaging parameters. This suggests that vaporised PCCAs have different acoustic behaviours from those of MBs. One possibility is that a change in the shell structure is responsible for the difference in behaviour between the precursor and vaporised bubbles. Mountford *et al* (2015) indicated that the lipid shell of PCCA is a 'bilayer' simply because it is formed by a condensing precursor MB, of which the 'monolayer' can be 'folded' after condensation. There is some evidence reporting that the lipid shell remained attached to the PCCA surface after condensing the MB (Mountford *et al* 2014), while the mechanism of how the lipid-shell structure affects PCCA vaporisation is still unknown. A vaporised PCCA may lose part of its lipid shell from its precursor MB counterpart, such that it could be less acoustically stable than its precursor MB. This might occur either when a MB is condensed into a nanodroplet, or, when a PCCA volumetrically expands back to a MB upon acoustic vaporisation, where the PCCA's 'bilayer' unzipped into the MB's 'monolayer'. Some studies have pointed out the dependence of the MB acoustic dissolution rate on phospholipid acyl chain length (Borden *et al* 2005), and that the stability of PCCA (e.g. circulation time) increases with a longer acyl chain (Sheeran *et al* 2017b), which may also provide a theoretical explanation for the signal decrease in the LFR imaging of the vaporised PCCAs.

For our experimental study, the ultrasound-driven coalescence, observed from the optics, may provide an initial explanation on the signal reductions for the first tens of milliseconds after activation. Postema *et al* (2004) shown that the MB coalescence event was observed to finish within several microseconds when the MBs were insonified by ten cycles of 0.5 MHz with high acoustic amplitudes (0.66–0.85 MPa). By comparison, we applied shorter but very rapid imaging pulses at a lower amplitude; the coalescence phenomena finished in a few milliseconds: 0.9 ms for the normal-sized vaporised PCCAs (see figure 8(i)) and 6.57 ms for the large ones was (see figure 8(ii)). This is probably due to the larger vaporised PCCAs oscillating less strongly than the smaller ones, and hence produced a smaller secondary Bjerkness force, which it took a longer time to coalesce (Mettin *et al* 1997). As the vaporised PCCAs coalesced, the population of the acoustic scatters (i.e. total MB numbers) decreased, as well as for the number of the vaporised PCCAs in a size comparable to that resonant with the ultrasound imaging driving frequency (i.e. 4 MHz). This could result in the decreasing trend for both the fundamental and harmonic signal. As for the B-mode signal, the vaporised native and size-selected PCCAs showed different characteristics in the later phase mainly because of the vaporised PCCA outliers. The vaporised PCCA outliers might undergo further growth in size (e.g. coalescence) exceeding a threshold point, after which the acoustic response at fundamental-frequency may dominate the B-mode signal. This is consistent with the results reported by Reznik *et al* (2011), in which there was an increase in the fundamental signal and simultaneous decrease in the harmonics after activation. Furthermore, the signal recover may be influenced by the properties of the host medium (Kang *et al* 2014), e.g. fluid viscosity and gas content, in which the diffusion of those gases may result in the vaporised PCCA's growth (Reznik *et al* 2012a, Shpak *et al* 2013). For the vaporised size-selected PCCAs, even after coalescence and 'clustering', they may not be able to provide a sufficient oscillation response, which is likely because of the mismatch between the ultrasound interrogation frequency and the resonance frequency of the vaporised PCCA population. Bader *et al* (2015) have indicated that coalescence via secondary radiation forces force bubbles to be in the resonant size, however the applied ultrasound parameters (i.e. peak-to-peak pressure of 0.44 MPa, continuous insonation period of 50 s) are very different from the lower-amplitude and short imaging pulses in this work. Future work is required to investigate the effect of imaging frequency on the vaporised PCCAs.

MB destruction, including shell rupture, counter diffusion (Sarkar *et al* 2009), and MB dissolution, which were not directly observed in this study, may also contribute to the disruption of the vaporised PCCAs, and thus result in the signal change. Stride and Saffari (2003) indicated that the destruction of the contrast agent could be expected when a high level of stress in the shell was maintained from a high-PRF insonation pressure. In our experimental setup, it was difficult to observe and determine such MB destruction phenomena, mainly because the displacement of the vaporised PCCAs driven by the ultrasound pulses and the out-of-focus effect as well as the limited frame-rate of the camera, was such that the disappearance of a vaporised PCCA cannot be conclusively determined by the destruction. In addition, a future study needs to investigate the relationship between the translation-speed/coalescence-time as well as the different sizes of the constrained vessels (Caskey *et al* 2006) with various ultrasound driving parameters (e.g. frequency, pulse length, amplitude, PRF). If the dynamics of the vaporised PCCAs can be controlled, it has potential implications for therapeutic applications, e.g. delivering a drug towards a tumour site with a desired translation-speed, creating tumour vascular occlusion with the coalesced bubbles with desired sizes.

We implemented a rapid and effective approach to remove the outliers in native PCCA emulsion while preserving the major sub-micron population (i.e. diameter <200 nm), which shows the potential of manipulating the population size for different diagnostic or therapeutic applications, e.g. microdroplets for embolization (Kripfgans *et al* 2005) and nanodroplets for extravasation (Kawabata *et al* 2005, Williams *et al* 2013). Other approaches have also been demonstrated to manipulate the size distribution of the stock PCCA solution, e.g. the

‘decantation’ method for nanodroplets (Sheeran *et al* 2016b) and ‘differential centrifugation’ for microdroplets (Mercado *et al* 2016).

Kripfgans *et al* (2005) reported a dosage for a different type of microdroplet solution that could result in adverse bio-effects for *in vivo* experiments, in which the concentration was at least ten times higher than that in this study. Kang *et al* (2014) also illustrated that the PCCA concentration within a certain range (applicable to this work) was independent from the temporal signal evolution (i.e. the change in the mean diameter of the vaporised PCCAs). Additionally, previous studies on MB coalescence have indicated that clustering effects of MBs happen when they are within a distance of a few radii (up to ten) as the interaction between MBs (secondary Bjerknes forces) decays as the inverse square of their distance (van der Meer *et al* 2007, Kooiman *et al* 2014). The PCCA concentration applied in this study is relatively low, resulting in an averaged spacing distance of 40.1 μm , and thus the distance between the vaporised PCCAs can be estimated to be 51-fold larger than its radius based on the theoretical and experimental evidence (Kripfgans *et al* 2000, Sheeran *et al* 2011, Shpak *et al* 2013) assuming a 200 nm PCCA vaporised into a 1 μm MB (an expansion factor of 5).

Future work is required to study the mechanism responsible for the evolution of vaporised PCCAs exposed to ultrasound interrogation. Also, a PCCA-specific imaging strategy could be developed to improve the detection of the vaporised PCCAs taking advantage of the fast changes in the temporary signals. Here we present initial evidence that the lipid-shelled PCCAs manufactured via ‘MB condensation’ were affected by the ultrasound imaging pulses. However, further investigation is needed for other types of PCCAs, especially for those with different encapsulation methods and gas core content. Sheeran *et al* (2012) indicated that the lifetime of vaporised PCCA can be controlled by its internal composition, e.g. perfluorocarbon compositions. On the other hand, it should be considered to improve the stability and lifetime of the vaporised PCCAs by optimising the PCCA manufacturing procedure. For instance, lung surfactant, Survanta, could be incorporated into the precursor MBs (Sirsi *et al* 2013) to increase lipid retention on the shell surface, with the hope of maintaining the MB to PCCA conversion efficiency and the PCCA activation threshold, while enhancing the persistence of the signal of the vaporised PCCAs. In addition, an *in vivo* situation should be considered in future experimental designs, e.g. flow condition. While a previous study (Kang *et al* 2014) has shown that the bubble dynamics (e.g. bubble growth, temporal signal change) was independent of flow velocity, a more complex physiological flow could potentially change the dynamics of the vaporised bubbles as they move out of the ultrasound field of view.

5. Conclusion

Compared with the MB contrast agents, the vaporised PCCA populations exhibit characteristic temporal signal evolutions with HFR ultrasound imaging, suggesting different acoustical behaviour between the MB and vaporised PCCA populations. It was also found that signals from the vaporised PCCA populations are significantly affected by the amplitude and frame rate of the HFR ultrasound imaging, as well as the size distribution of the PCCAs. A change in the size (e.g. ultrasound-induced coalescence) of the vaporised PCCA populations was observed from the simultaneous optical study. This experimental finding provides more insights into PCCA behaviour and informs future development of PCCA-associated imaging strategies.

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Conflict of interest

The authors declare that there are no conflicts of interest.

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